

E-COMMERCE REVENUE OPTIMISATION & BUSINESS PERFORMANCE ANALYSIS

Business Analytics Project using Python & Exploratory Data Analysis

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Tools Used

Python | Pandas | NumPy | Matplotlib | Seaborn

April 2026

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1. PROJECT BACKGROUND AND CONTEXT

The e-commerce market has witnessed a high growth rate in the last ten years with the growth in internet use, improvement of internet payment systems and the changing consumer tastes and preferences. The current online retail systems produce enormous amounts of transactional data every day, with finer details on customer purchases, product demand, pricing and geographic distribution of sales. This information forms a great business asset, since it can give profound information on how businesses are performing and how customers behave when effectively analysed.

Although such rich datasets are available, most organizations find it hard to access this information to the fullest extent in making decisions. Practically, intuition or previous experience is frequently used to make business decisions connected with *product strategy*, *targeting customers*, and *inventory planning*, instead of analyzing data systematically. This mismatch between the data available and its efficient utilization underlines the increased prominence of **business analytics** in the contemporary e-commerce context. When organizations effectively utilize data, they are more likely to discover the trends, maximize resources and improve overall business performance.

This project aims to look at transactional data on an online retail dataset using the data to gain insight into how revenue is created and distributed on various levels like *products*, *customers*, and *geographic regions*. The data includes specifics of transactions, such as invoice numbers, product descriptions, purchased quantities, unit prices, customer identifiers, and information about the country. The dataset is a complete depiction of the real-world e-commerce activity and offers a valuable insight into business performance with more than **541, 000 transaction records**.

The main issue that was discussed in this project is the absence of understanding of the main drivers of revenue in an e-commerce company. It is not always obvious in most instances which products are the ones that bring in the most income, which clients bring about the best value, and what markets are the most important in terms of sales. Lack of such insights can result in businesses investing resources in an inefficient manner, failing to pursue growth opportunities, or neglecting high-impact areas. Analyzing transaction-level data is thus needed to determine patterns that can be used to make more strategic decisions and enhance operational performance.

The project uses data analytics methods to examine and analyze trends in the dataset to win this issue. The analysis is concentrated on a number of key factors such as *revenue allocation*, *seasonal performance*, *product performance*, *customer contribution* and *geographic sales*. Through these dimensions, the project will create actionable outputs that can help to **optimize revenues** and make better business decisions.

The data in this project was sourced on Kaggle and is a reflection of actual transactional data in an online retail company that is based in more than one country. The data covers **38 countries** transactions and enables analysis of the distribution of sales geographically and the market concentration. This global presence allows locating the areas that can bring the greatest part of total revenues and where there can be more growth opportunities.

Dataset Source: <https://www.kaggle.com/datasets/carrie1/ecommerce-data>

The other important element of the project is the establishment of a **Revenue variable**, which is obtained by the product of the *number of products sold* and the *unit price of the products*. This variable is the basis of all the other analyses since it gives a direct measure of the value produced by each transaction. The project can assess business performance in a consistent, clear and measurable way by emphasizing on the revenue as the core metric.

The project will attempt to provide some key business questions through this method of analysis. These are the identification of the products that contribute the most to revenue, the knowledge of the way the revenue varies with time, the assessment of the customers who produce the most value and the distribution of the sales across the various countries. Also, the analysis explores if revenue is concentrated on a few products or customers and this has significant strategic implications on the decision-making process and resource allocation.

The answers to these questions are necessary in order to enhance decision making in an e-commerce situation. As an example, determining products that are performing well can assist companies to focus on marketing and inventory control. Knowledge of customer contribution is useful in the formulation of specific retention strategies and geographic distribution may help in expansion into new markets. On the same note, seasonal trends can be identified to enable businesses make better plans on inventory levels, promotions and supply chain operations.

Moreover, the project emphasizes the use of data-driven insights in minimizing uncertainties in business decisions. Organizations can use analytics to make informed decisions that are informed rather than basing them on assumptions, which is supported by actual transaction data. This not only enhances efficiency but also increases the capability of responding to the ever-evolving market conditions and customer preferences.

On the whole, the project shows the potential of **data analytics to be used in real business issues in the field of e-commerce**. Through the analysis of raw transactional data into meaningful insights, the study shows the importance of analytics in improving business insight and aiding in the process of making data-driven decisions. The conclusions of this project could assist the businesses to maximize their revenue, to be more efficient and to create better strategies in a more competitive digital market.

2. OBJECTIVES AND SCOPE

2.1 OBJECTIVES

The main aim of the present project is to examine transactional data of an e-commerce dataset to gain insight into some of the major drivers of revenue and business performance. The research is dedicated to deriving significant information to aid in the decision-making process based on data in an online retail setup.

The project has the following specific objectives:

- To examine **trends and distributions of revenues** of the transactions in order to comprehend the way revenue is generated in the business.
- To determine the **best selling products** and determine their contribution to the total sales.
- To study the **demand trend of products** in terms of quantity sold and buying behaviour of the customer.

- To assess the **country-wise revenue distribution** and define the most important geographic markets that underlie the sales.
- To examine the amount of **customer contribution to the total revenue** and determine high value customers.
- To determine the **revenue concentration based on the Pareto principle (80/20 rule)**
- To extract **business knowledge and suggestions** that can aid in revenue optimization strategies.

These goals will make sure that the analysis is focused on business practical use and produce actionable results.

2.2 SCOPE OF THE STUDY

This project will focus on the historical transactional data analysis through descriptive analytics methods. The analysis concentrates on the knowledge of the previous patterns and trends and not foreseeing what is going to happen.

Included in Scope:

- Analysis based on revenue of transactional data.
- The transactional data may be analyzed by revenue-based analysis.
- Product performance evaluation
- Customer-level contribution analysis
- Geographic (country-wise) sales analysis
- The identification of the pattern of revenue concentration
- Development of business insights and recommendations.

Excluded from Scope:

- Profitability or cost analysis, since the data on costs is not accessible in the data.
- Predictive modelling or machine learning techniques
- Hypothesis testing or sophisticated statistical analysis.
- External market forces like competition or macroeconomic forces.

3. PROJECT WORKFLOW

All the analysis steps (data preprocessing, feature engineering, and the complete implementation) of this project were implemented in Python in Google Colab. The entire notebook including code, outputs, and visualizations are available by the following link:

Project Notebook:

https://colab.research.google.com/drive/1CVM_R_658tNJq_1qzuvHHIM1Xi1HD19H?usp=sharing

3.1 DATA COLLECTION AND DATASET OVERVIEW

The initial proposal in this project was to find and acquire an appropriate dataset to analyse the business performance within the e-commerce industry. A dataset that is publicly available in Kaggle (transactions of an online retail company) was chosen. The data set contains the information on customer purchases, product description, quantity, price, the date of the invoice, and geographic location.

The dataset used is of **541,909 transaction records and 8 variables** which is large enough to carry out meaningful analysis. It contains important variables which include *InvoiceNo*, *StockCode*, *Description*, *Quantity*, *InvoiceDate*, *UnitPrice*, *CustomerID* and *Country*. These variables form the basis of the revenue patterns, customer behaviour and geographic distribution of sales analysis.

The Pandas library was used to import the data into Python, and preliminary exploration was carried out in order to learn more about its structure and content.

FIGURE 1: PREVIEW OF RAW DATA (FIRST 5 ROWS)

```
# LOADING THE E-COMMERCE DATASET
df = pd.read_csv("OnlineRetail_Ecommerce_Dataset.csv", encoding="ISO-8859-1")

# DISPLAYING FIRST 5 ROWS
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12-01-2010 08:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12-01-2010 08:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12-01-2010 08:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12-01-2010 08:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12-01-2010 08:26	3.39	17850.0	United Kingdom

OBSERVATION

- The data contains several records of each invoice, each row shows product in a transaction.
- Prices, quantities and product descriptions are easily visible.
- Some transactions involve several items meaning that one invoice can consist of several products.

INTERPRETATION

This format shows that the data that is being captured reflects **transaction level information at a fine level**, and thus, purchasing behaviour can be analyzed in detail. As every row corresponds to a single product in an order, the total revenue per transaction can be calculated and the performance of specific products analysed.

Additional data analysis of the dataset format was performed to analyze the type of data and find missing values.

FIGURE 2: DATA STRUCTURE AND INFORMATION.

```
# CHECKING DATA TYPES & MISSING VALUES
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   InvoiceNo    541909 non-null object
1   StockCode    541909 non-null object
2   Description  540455 non-null object
3   Quantity     541909 non-null int64
4   InvoiceDate   541909 non-null object
5   UnitPrice    541909 non-null float64
6   CustomerID   406829 non-null float64
7   Country      541909 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

OBSERVATION

- There are **541,909 entries** in the dataset in **8 columns**.
- The *CustomerID* column has blank values.
- The *InvoiceDate* variable will be initially of object data type.
- Quantity and UnitPrice are numbers that can be calculated.

INTERPRETATION

The fact that some of the values are missing in the *CustomerID* column is an indication that not every transaction has identifiable customers. This involves implications on customer level analysis and must be properly addressed during data cleaning. Also, the *InvoiceDate* column should be converted to a datetime type to be able to analyze it by time, including monthly revenue patterns.

3.2 DATA CLEANING AND PREPROCESSING

Once the data structure of the dataset was familiar, cleaning and preprocessing the data followed in order to assure accuracy and reliability of the analysis. Raw transactional data sets usually have irregularities like a lack of values, invalid entries, and improper formats, which may adversely affect the outcomes of the analytical process, unless they are managed appropriately.

In this dataset there are various data quality problems which were found out in the first exploration stage. These were cancelled transactions, negative quantities, missing customer identifiers and wrong data types. As such, a set of preprocessing techniques were used to make the dataset ready to be analyzed.

The algorithm to clean the data was done in Python in Google Colab, and the major steps are shown below.

FIGURE 3: DATA CLEANING PROCESS (CODE SNIPPET).

```
# REMOVING CANCELLED ORDERS (INVOICE STARTING WITH 'C')
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]

# REMOVING NEGATIVE QUANTITIES
df = df[df['Quantity'] > 0]

# REMOVING MISSING CUSTOMER IDS
df = df.dropna(subset=['CustomerID'])

# CONVERTING DATA COLUMN TO DATETIME
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

STEPS PERFORMED

- **Cancellation of Transactions:** Cancelled orders are those that have the invoice numbers beginning with the letter “C”. These have been eliminated so as to only analyse completed transactions.
- **Elimination of Negative quantities:** The negative figures in the *Quantity* column are product returns. These entries were not included since the aim of the analysis is to assess the revenue generation.

- **Missing Values:** There were missing values in the *CustomerID* column. Such rows were eliminated in order to facilitate a proper customer level analysis.
- **Date Format Change:** The *InvoiceDate* variable was changed to an object type to date format to allow conducting time-based analysis like monthly trends.

OBSERVATION

- Many rows were deleted because of lack of *CustomerID* values.
- There were cancelled and return transactions in the dataset.
- The data needed to be cleaned and then some meaningful analysis could be done.

INTERPRETATION

The occurrence of cancelled transactions and negative quantities imply that raw dataset has both successful and reversed transactions. This would be misleading and would result in the miscalculation of revenues by including such entries. Filtering of such records, therefore, makes sure that the analysis represents real sales performance.

Likewise, deletion of rows that contain no *CustomerID* would enhance the accuracy of the customer-level analysis. It would not be possible to effectively measure customer contribution or purchasing behaviour without unique customer identifiers.

By transforming the *InvoiceDate* column to a datetime format, time-series analysis is possible, which is crucial in determining seasonal trends and a time-based pattern of revenues.

3.3 FEATURE ENGINEERING

Once the data had been cleaned and preprocessed, the next thing to do was to generate new variables that would aid in further analysis. This is referred to as **feature engineering** and it is significant in converting raw data into significant metrics to be analyzed.

A new variable was generated in this project namely **Revenue**. This variable was determined by multiplying the *Quantity* of products bought by the *UnitPrice* of each product:

$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$

This was calculated to every transaction in the dataset, so that the total value created by every purchase could be calculated.

FIGURE 4: FEATURE ENGINEERING - CREATION OF REVENUE VARIABLE

```
# CREATING REVENUE COLUMN
df['Revenue'] = df['Quantity'] * df['UnitPrice']

df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

OBSERVATION

- Another column called *Revenue* was successfully added to the dataset.
- The rows now reflect the value of the summation of that line of transaction.

- Revenue values are quite different in various transactions, which rely on quantity and price.

INTERPRETATION

The development of the *Revenue* variable is an important aspect of the analysis, because it will give a direct indication of the performance of the business. Rather than looking at the quantity and price separately, a combination of the two into one metric can be used so as to make a better analysis of the level of value add to the business by each transaction.

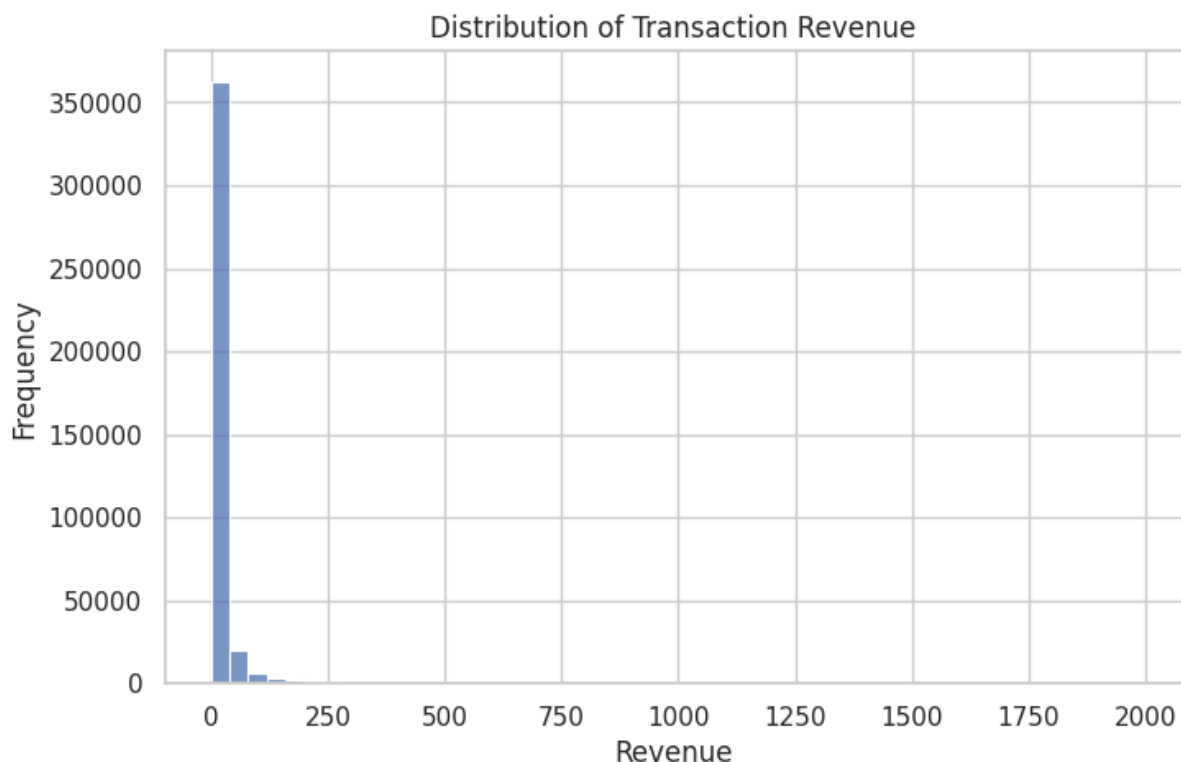
The variable will be the premise of all other analyses that may be conducted in the project, such as distribution of revenues, performance of the product, customer contribution, and geographical patterns of sales. The analysis will be more in line with the business goals as revenue is one of the most important financial performance indicators.

3.4 REVENUE DISTRIBUTION ANALYSIS

Once the *Revenue* variable was created, it was followed by the analysis of how the revenue is spread out in transactions. Knowing the revenue distribution is valuable since it can determine whether the business has regular revenue generation per order or high-value transactions.

A histogram was used to analyse this with the *Revenue* variable. Outliers were clustered in order to be filtered to visualization in order to allow the distribution to be read more easily.

FIGURE 5: TRANSACTION REVENUE DISTRIBUTION (HISTOGRAM).



OBSERVATION

- Many of the transactions have a relatively low value of revenue most of which are **below 100**.
- Few transactions bring in a lot more revenue.

- The dispersion is extremely skewed on the right.
- There are very few transactions which have very high revenue in comparison to the majority.

INTERPRETATION

The distribution of revenue is **skewed right**, which is the typical distribution of retail and e-commerce data. This means that there is a high amount of small value transactions in the business, and fewer high-value transactions.

This trend indicates that the individual transactions might not yield a lot of revenue but when the sum of a number of small purchases is taken into consideration, it adds up to a lot of sales. Concurrently, high-value transactions suggest that there are premium sales or bulk purchases.

In business terms, such an insight illustrates the need to strike the right balance between both types of transactions. *Product bundling*, *cross-selling*, and *promotions targeted* at smaller transactions can be used to enhance the value of smaller transactions and hence the overall revenue performance.

Also, the fact that the high-value transactions may be present implies that some customers or products can bring revenue in disproportion. Such an insight will be used in the subsequent analysis in later sections, especially in determining the high-value customers and the best performing products.

3.5 MONTHLY REVENUE TREND ANALYSIS

Once the distribution of revenue among transactions was known, the next thing to do was to analyse the distribution of the revenue as a time-varying variable. Time analysis is crucial in e-commerce because it aids in determining **seasonal cycles**, **demand variations** and **peak business seasons**, which is essential in business planning and decision making.

In order to achieve this analysis, the *InvoiceDate* variable was transformed to a datetime format during the preprocessing process. The data was further broken down by months and the total revenue per month was estimated. This enabled it to identify trends in the generation of revenue over the years.

FIGURE 6: MONTHLY REVENUE TREND.



OBSERVATION

- Monthly revenue is around between **450,000 and 1,160,000**.
- In the first months of the year, revenue does not decrease significantly.
- A noticeable increase in revenue is observed from **September onwards**
- The most revenue is registered in **November**, which is over **1,100,000**.
- There is a decrease in revenue in December.

INTERPRETATION

The trend of monthly revenue shows that there are certainly **seasonal trends** within the data set. The fact that the revenue saw a remarkable growth in the later months of the year and especially in November indicates that the business is in high demand during the holiday period or other large-scale shopping occasions.

This trend aligns with the usual e-commerce behaviour whereby sales are boosted around the festivals, promotional events, and during end of year shopping activities. The high of November can be linked to other events like the massive sales of discounts or holiday shopping habits.

This is because the drop in December may not be a decrease in sales performance, but could be an incomplete data of December. This shows the value of a cautious interpretation of trends and taking into account data limitations.

In the business context, the analysis is useful in terms of **inventory management, marketing and resource management**. Companies are also able to have higher stock levels, organize promotional activities, and streamline logistic processes during the peak months in order to maximize profits. Likewise, the awareness of low-demand times can be used to develop cost-cutting measures and specific advertisement campaigns.

3.6 PRODUCT PERFORMANCE ANALYSIS

Knowledge about product performance is critical in determining the products that bring about the greatest contribution to the total revenue and also the most sought out products. A limited set of products in an e-commerce business can be a very significant portion of total sales and hence, the product level analysis is an important part of the decision-making process.

Two important methods were employed to analyse the product performance in this project:

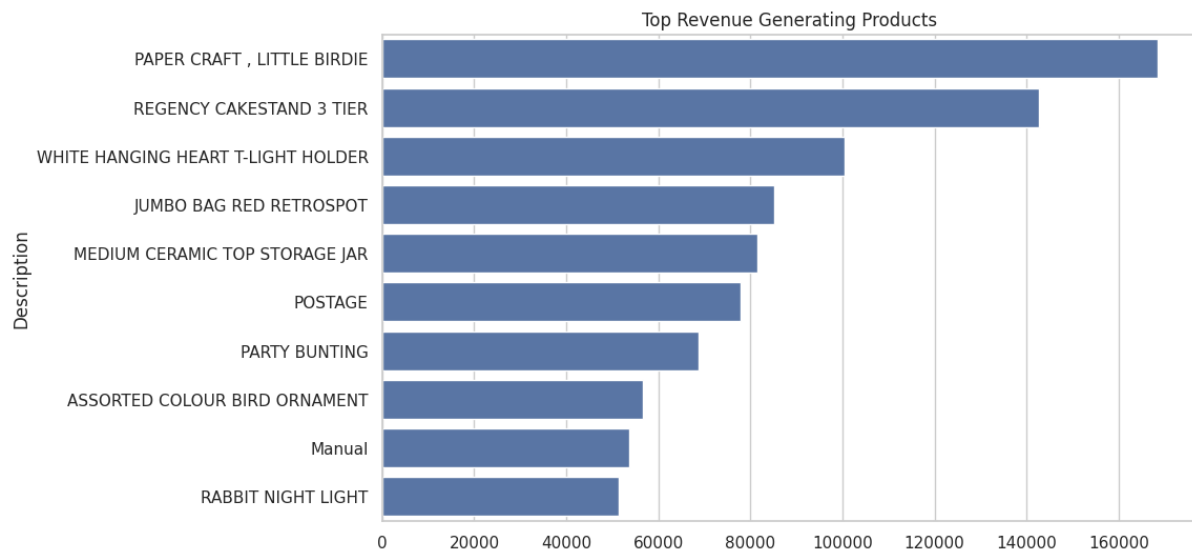
1. **Revenue-based analysis**
2. **Quantity-based analysis**

These two views aid in the realization of **value created** and the **products demand**.

3.6.1 TOP REVENUE GENERATING PRODUCTS

The dataset was sorted by product description in order to find the products that generated the highest amount of revenue and the amount of revenue was totaled per product. The top 10 products were then chosen based on the contribution to the revenue.

FIGURE 7: TOP REVENUE GENERATING PRODUCTS.



OBSERVATION

- The “PAPER CRAFT, LITTLE BIRDIE” product is the one that brings the most revenue, more than 160,000.
- Top 10 products have a large share of total revenue.
- There is an observable difference between the top product and others.

INTERPRETATION

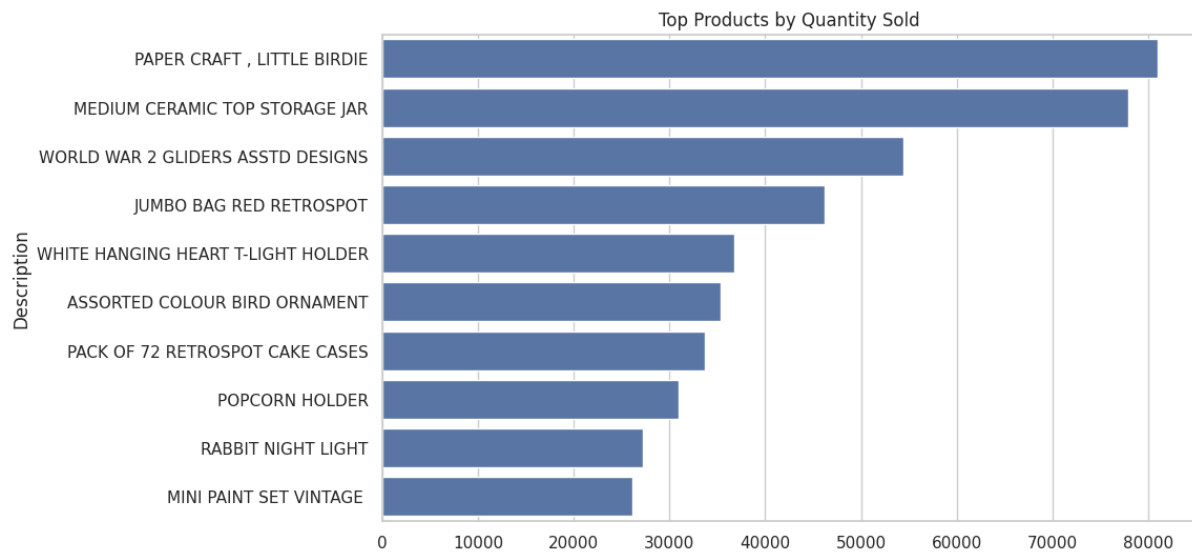
The analysis shows that revenue is **extremely concentrated on a few products**. Domination of the best product implies that some products are significant in the business performance.

This focus demonstrates the need to designate and focus on high-performing products. Companies can target their marketing activities, promotions and their inventory management to these products in order to maximize the revenue. Concurrently, there is a risk of being dependent on few products, where any reduction in the demand of these products would greatly affect the total sales.

3.6.2 TOP PRODUCTS BY QUANTITY SOLD

In order to determine the product demand, the dataset was analysed according to the total quantity sold of the product. Top 10 products were determined by the sales units.

FIGURE 8: TOP PRODUCTS BY QUANTITY SOLD.



OBSERVATION

- The volume of sales of some products is extremely high, over **80,000 units sold**.
- The highest revenue generating products are not necessarily high demand products.
- There is discrepancy between quantity-based ranking and revenue-based ranking.

INTERPRETATION

This discussion shows that the **high demand may not always translate into a high revenue**. Certain products can be sold in volume and at low prices whereas the other ones are able to make more revenue at a low volume of sales.

This is a significant difference in terms of business. High-volume products might be a necessity to ensure customer activity and steady sales and gain, but high-revenue products have a greater impact on profitability.

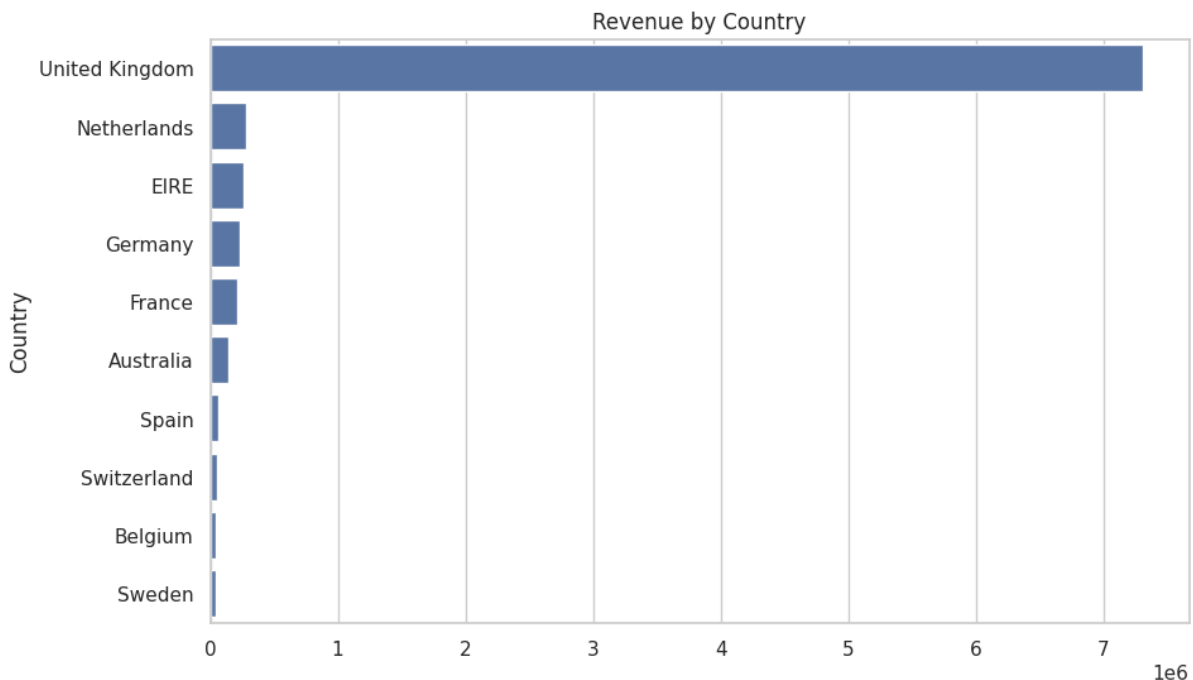
Companies need to pursue a moderate approach, both by making high-demand products available and by marketing high-value products. These strategies can be combined to optimize the sales volume and revenue creation.

3.7 COUNTRY-WISE REVENUE ANALYSIS

Once the performance of the products has been analysed, the second activity was to consider the distribution of the revenue to the various countries. Geographic analysis plays a significant role in e-commerce because it assists in determining the prime markets, distribution of customers, as well as the level of market concentration.

In order to conduct this analysis, the data was categorized using the *Country* variable and the amount of total revenue was determined per country. A bar chart was then used to visualize the results to compare the revenue contributions in various regions.

FIGURE 9: REVENUE BY COUNTRY.



OBSERVATION

- The **United Kingdom** produces the greatest revenue of more than **7 million**.
- Other nations contribute very little revenue as compared to the UK.
- There is sharp decline following the top country, which is a sign of uneven distribution.
- Few countries make significant contributions to overall income.

INTERPRETATION

It is apparent in the analysis that the level of the **concentration of the revenue is highly placed in one geographical region** with the United Kingdom prevailing over the overall sales. This means that the enterprise depends on a single market to generate revenues.

There are opportunities and threats in such concentration. On the one hand, good results in one of the major markets imply a well-developed customer base and stable demand. Conversely, excessive reliance on one region will raise the exposure to market fluctuations, economic variations or competition in the region.

Strategically, this insight implies that there is a need to **geographically diversify**. Diversification to other markets and increasing international presence can serve to mitigate risk and generate new growth opportunities. Moreover, specific marketing plans could be created to allocate underperforming regions to contribute more towards the overall revenue.

3.8 CUSTOMER CONTRIBUTION ANALYSIS

Knowledge on customer contribution is needed to assess the distribution of revenue among the customer base. A few customers bring in high percentage of overall revenue in most business and as such, it is necessary to determine and examine these high-value customers.

In order to conduct this analysis, the data was clustered by *CustomerID* and the total revenue generated by the individual customer was determined. The best customers were then determined on the basis of their contribution to total revenue.

FIGURE 10: TOP CUSTOMERS BY REVENUE (TABLE OUTPUT).

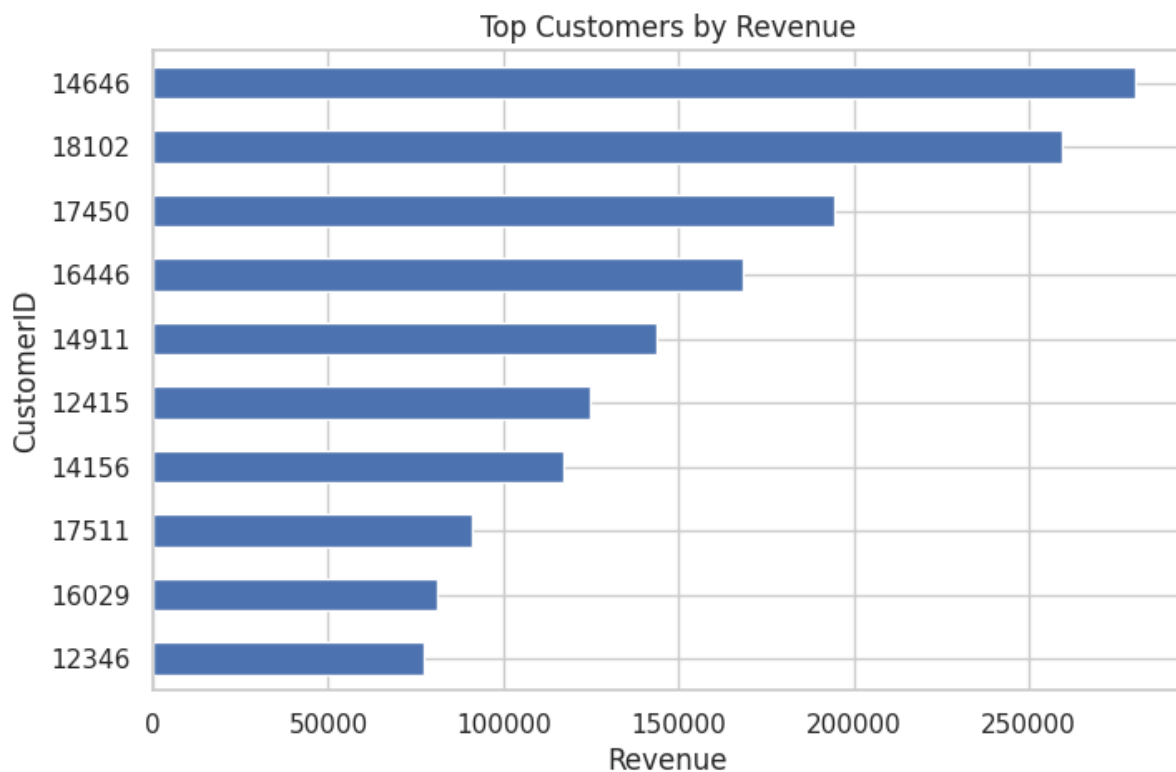
```
# CALCULATING TOTAL REVENUE GENERATED BY EACH CUSTOMER
customer_revenue = df.groupby('CustomerID')['Revenue'].sum().sort_values(ascending=False).head(10)

customer_revenue
```

CustomerID	Revenue
14646.0	280206.02
18102.0	259657.30
17450.0	194550.79
16446.0	168472.50
14911.0	143825.06
12415.0	124914.53
14156.0	117379.63
17511.0	91062.38
16029.0	81024.84
12346.0	77183.60

dtype: float64

FIGURE 11: TOP CUSTOMERS BY REVENUE (BAR CHART)



OBSERVATION

- The best customer has a revenue of about **280,000**.
- Other high customers give an input of **75,000-200,000**.
- High-value customers are vastly different, unlike their counterparts.
- There is a small number of customers who contribute a proportionately large portion of total revenue.

INTERPRETATION

The analysis shows that the **revenue is highly skewed with few customers**, which is a common trend in e-commerce companies. This implies that a limited number of customers are vital in the overall business performance.

As a business, these premium customers are invaluable and must be targeted by using special strategies. Businesses can implement **loyalty programs, personalized offers, and exclusive promotions** to retain these customers and enhance their lifetime value.

Simultaneously, the lack of diversification can pose threats due to over-dependence on a limited number of clients. When such customers cut down on their expenditure or quit purchasing, this may affect revenue to a great extent. As such, it is also necessary to increase the customer base and turn medium-value customers to high-value customers.

3.9 AVERAGE ORDER VALUE (AOV) ANALYSIS

Having analysed customer contribution, the second step was to analyze the **Average Order Value (AOV)** next. AOV is a significant business indicator that estimates the average revenue per order. It gives information on customer buying behaviour and assists companies to know the level of value created with each transaction.

In calculating AOV, the total revenue has been divided by the total amount of unique invoices. This will make sure that the measure captures the average value per order and not per product.

FIGURE 12: AOV CALCULATION OUTPUT

```
# CALCULATING AVERAGE ORDER VALUES
total_revenue = df['Revenue'].sum()
total_orders = df['InvoiceNo'].nunique()

AOV = total_revenue / total_orders

print("Average Order Value:", round(AOV,2))

Average Order Value: 480.76
```

OBSERVATION

- The Mean OV is about **480.76**.
- Every transaction will produce an **average of 480 units of revenue**.
- There is a variation in transactions, but the average is quite moderate.

INTERPRETATION

The AOV value implies that customers usually buy more than one item or more valuable goods at once. This indicates that the business can yield a decent level of revenue per order which is good sign of sales performance.

In business terms, AOV is a very important measurement since it can be directly used to enhance the overall revenue without necessarily the number of customers. *Product bundling, cross selling and promotional offers* are some of the strategies that can be used to encourage customers to purchase more every time they transact.

Moreover, AOV analysis combined with customer and product data can assist companies in finding the ways to improve customer experience and optimise the value of purchases.

3.10 REVENUE CONCENTRATION ANALYSIS (PARETO ANALYSIS)

Once the Average Order Value has been analysed, the next analysis was to understand the distribution of revenue among products. In most companies, a few products bear a significant part of the overall income. The **Pareto Principle**, also referred to as the *80/20 rule*, is the explanation of this concept: about 80 percent of the outcomes are caused by 20 percent of the reasons.

To interpolate this, the products were ranked in terms of the total contribution of revenue. This was followed by the cumulative revenue which was computed in order to know the contribution of the best performing products.

FIGURE 13: REVENUE CONTRIBUTION OF TOP 20% PRODUCTS.

```
# ANALYZING REVENUE CONCENTRATION
product_revenue = df.groupby('Description')['Revenue'].sum().sort_values(ascending=False)

top_20_percent = product_revenue.head(int(len(product_revenue)*0.2))

print("Revenue from top 20% products:", round(top_20_percent.sum(),2))

Revenue from top 20% products: 7036402.86
```

OBSERVATION

- The top **20 percent of the products produce around 7,036,402.86 in revenue.**
- A small proportion of total sales is contributed by a relatively small number of products.
- Distribution of revenue is extremely skewed among the products.

INTERPRETATION

The findings are clear evidence of the existence of the **Pareto Principle** in the data. Few commodities drive a large share of overall revenue, with most commodities having relatively small contributions.

This is a valuable insight, in terms of business. It emphasizes the need to identify and concentrate on best performing products since they are significant in revenue generation. These products can be given the first priority by businesses in terms of marketing, inventory control, and promotion strategies to get the best returns.

Meanwhile, the analysis also implies that there are a lot of products that bring in a very minimal portion of revenue. This gives a business a chance to assess their product line, drop nonperforming products or enhance their business by enhancing position and marketing.

4. TOOLS AND FRAMEWORKS USED

4.1 TOOLS USED

In this project, the analysis was done with a combination of programming tools and data analysis libraries that facilitated efficient processing, visualization and interpretation of data.

- **Python:** Python was the main programming language to be employed in data analysis since it is simple, flexible and has a wide library to support data analytics activities.

- **Google Colab:** The code was executed in Google Colab. It offers a cloud based platform which enables easy access of the computing resources and does not need local installation. It also works with interactive notebooks, which allows it to be easy to mix code, outputs and explanations.
- **Pandas:** The data manipulation and analysis were performed with the help of the Pandas library. It facilitated the management of vast amounts of data, including data cleaning, grouping, filtering and aggregation.
- **NumPy:** NumPy was employed to perform numerical operations and calculations especially with array manipulation and mathematical calculations.
- **Matplotlib and Seaborn:** Data were visualized with the help of these libraries. The fundamental plotting capability was supplied by Matplotlib, and to produce more informative and visually engaging charts (histograms, line graphs, bar charts, etc.) Seaborn was utilized.

4.2 ANALYTICAL FRAMEWORKS

Along with tools, the project utilized a number of approaches and frameworks of the analysis to draw meaningful conclusions based on the dataset.

- **Exploratory Data Analysis (EDA):** EDA was employed to get an idea of the data structure, patterns and detect anomalies and to get some general insights about the data prior to the in-depth analysis.
- **Descriptive Analytics:** The project is based mostly on descriptive analytics that aims to examine previous data to comprehend the past data and performance. This will be useful in establishing some main trends and summarization of data meaningfully.
- **Pareto Principle (80/20 Rule):** Pareto Principle was employed to analyze the revenue concentration. It assisted in determining whether a small percentage of the products would generate a high percentage of the total revenue.

In general, all these tools and frameworks allowed conducting efficient data processing, meaningful interpretation, and effective visualization. This guaranteed the analysis was organized, sound and business oriented.

5. OUTCOMES / DELIVERABLES

The project was able to apply data analytics methods to an e-commerce transactional dataset providing a series of useful outputs that blend technical implementation and business information.

5.1 DATA PREPARATION AND PROCESSING

- A **cleaned and organized data** was created by eliminating cancelled transactions, negative quantities and incomplete customer identifiers.
- Data were converted into a manageable format to be analyzed, making it accurate and consistent.
- The data was prepared to do a revenue-based and customer-level analysis.

5.2 FEATURE ENGINEERING

- A new variable **Revenue** was created using:
$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$
- It was able to directly measure the value of transactions.
- The Revenue variable was made the **centre** of all further analysis.

5.3 DATA VISUALIZATION

The project generated several visualizations to depict some major patterns and trends:

- **Revenue distribution histogram** to get the feel of the patterns of transaction values.
- **Monthly revenue trend line chart**: To determine seasonality.
- **Product performance** (quantity and revenue) **bar charts**.
- The **country-wise revenue chart** to examine the geographic distribution.
- **Customer contribution charts** in order to know high value customers.

Such visualizations enhanced the interpretability and simplified the understanding of complex data.

5.4 ANALYTICAL INSIGHTS

The analysis led to some major findings:

- The products with **high concentration of revenue** are few.
- The **United Kingdom has the highest contribution to revenue**.
- There exists a narrow segment of **premium clientele with a large share in the total revenue**.
- Revenue: There is **seasonal variation** in revenue with peaks in the later months of the year.
- The Average Order Value (~480.76) is a moderate customer spending per transaction.

5.5 BUSINESS RECOMMENDATIONS

According to the analysis, actionable recommendations of the project were:

- Target **best selling products** in order to maximize revenues.
- Develop **customer retention strategies** of high valued customers.
- Grow AOV by means of **bundling and cross-selling**.
- **Internationalize** to decrease geographic dependence.
- Match inventory planning and **seasonal demand patterns**.

5.6 OVERALL OUTCOME

On the whole, the project indicates how well data analytics can be used to convert raw transactional data into **business insights that can be taken into action**. The deliverables are structured data, visual analysis, and strategic recommendations that can be used to enhance best decision-making in an e-commerce setup.

6. KEY PROJECT LEARNINGS

The initial stage of this project was a great experience in practice of using the data analytics methods on a business dataset. The involvement in working with a big e-commerce transactional dataset also assisted in the creation of a more profound perspective of how raw data can be refined into useful insights that can be used in business decision making.

6.1 TECHNICAL LEARNING

One of the key learnings from this project was the practical application of **Python for data analysis**. With libraries like *Pandas* and *NumPy*, I easily cleaned, filtered, grouped and aggregated data. This enhanced my knowledge on the possible way of processing and managing data of large size in a systematic way.

I also got acquainted with working with **data visualization** through *Matplotlib* and *Seaborn*. The use of charts, including histograms, line charts, and bar charts, served to learn more about how visual representation can make complex information easier to comprehend and the patterns easier to interpret.

One more valuable technical learning was the principle of **feature engineering**, especially, the development of the *Revenue* variable. This step demonstrated the ability of using existing variables to build more meaningful measures to examine.

6.2 UNDERSTANDING OF DATA CLEANING AND PREPARATION

The project highlighted the fact that it is essential to **pre-process data** prior to any analysis. Managing problems of missing values, negative quantities and cancelled transactions made me realize that raw datasets usually have discrepancies that have to be resolved, to get the correct outcomes.

The process is what enhanced my ability to be attentive to details and to the fact that the quality of analysis is greatly reliant on the quality of data. It also emphasized the significance of making rational choices in the course of cleaning e.g. eliminating irrelevant or misleading records.

6.3 BUSINESS AND ANALYTICAL UNDERSTANDING

In addition to technical skills, the project allowed me to learn a lot about the **concepts of business analytics**. I learned how to interpret the data in business as opposed to seeing numbers. For example:

- Examining the performance of the products assisted in the realisation of how it is possible to focus the revenue on a limited number of products.
- Customer contribution analysis indicated the significance of high-value customers.
- Country-wise analysis showed that geographic concentration could be a factor affecting business risk.

These observations assisted in closing the data analysis-business decision gap.

6.4 PROBLEM-SOLVING AND ANALYTICAL THINKING

As part of the project, I have become more adept at **solving problems** as I was able to recognize problems within the dataset and figure out how they could be overcome. The choice that needed to be made in terms of cleaning the data and the variables to concentrate on and the way to organize the analysis needed logical thinking and analytical reasoning.

Another skill that the project enhanced is the critical interpretation of results. I have learned to enquire *why there are patterns* and what they mean to the business rather than just observe them.

6.5 PRACTICAL EXPOSURE TO REAL-WORLD DATA

Using a real-life data set was helpful in getting experience on how data is organized and applied in real world business situations. This project was different as opposed to theoretical exercises since it involved the handling of incomplete, inconsistent and unstructured data, a common scenario in the real-life situations.

The experience allowed me to learn that data analysis is not as straightforward as it may seem, and flexibility during work with real data is crucial.

6.6 OVERALL REFLECTION

On the whole, this project reinforced my knowledge of technical and business elements of data analytics. It boosted my data analysis skills, interpretation, and inference of action.

The project also helped me to gain more confidence in the use of analytical tools and their application to real world problems. It offered a real-life basis on how businesses can utilize data in enhancing performance and making sound decisions.

7. MANAGERIAL IMPLICATIONS

The results of the analysis of the e-commerce data sets have significant implications on business strategy and decision making. These implications show how organizations can enhance performance in terms of revenues, maximize operations, and minimize possible risks through a structured, data-based approach. Businesses can go beyond decisions about things they need to make because of intuition and concentrate on the numbers by translating the results of analysis into strategies that can be implemented.

7.1 PRODUCT STRATEGY

The analysis of the product performance revealed **high concentration of revenue in few products**, which means that a small number of items are playing a key role towards overall business performance. The concentration indicates that not every product has the same contribution and hence there is need to focus on strategies.

Recommended Actions:

- Create **awareness and greater exposure** of high performing products using specific marketing campaigns.
- Make sure that there is **regular inventory** of high-scarce products to avoid stockouts.
- Implement high-revenue product **premium positioning/bundling**.

- Examine and streamline **poor performing products** through change in pricing, packaging or marketing.

Expected Impact: Diverting towards high performing products can greatly enhance revenue and reduce inefficiencies. Meanwhile, it is possible to optimize the operations and enhance profitability by enhancing or removing products with poor performance.

7.2 CUSTOMER STRATEGY

The customer contribution analysis indicated that a **few customers are contributing relatively excessive proportion of total revenue**, thus the need to identify and retain such high-value customers.

Recommended Actions:

- Create **loyalty programs** to reward repeat purchases and a long-term effort.
- Offer **customized suggestions and offers** according to the buying history.
- Provide **special treatment**, including priority to sales or superior services.
- Institute measures to **transform medium value customers into high-value customers** like special offers.

Potential Impact: These measures will be able to enhance customer retention, customer lifetime value, and a more stable revenue base. Simultaneously, increasing the high value customers will decrease reliance on a few.

7.3 MARKETING AND SALES STRATEGY

The revenue distribution and **Average Order Value (AOV $\approx$ 480.76)** analysis indicates that the customers add moderate value to the transaction but there is an opportunity to significantly increase the amount of spending per order.

Recommended Actions:

- **Product bundle** to motivate customers towards purchasing more than one item.
- Use **cross selling and upselling strategies** in the buying process.
- Develop **seasonal and event-based marketing programs** and times of maximum demand.
- Create **temporary promotions and savings** to create a sense of urgency in purchasing.

Projected Impact: These measures will be able to boost the Average Order Value and total revenue without necessarily having to boost the number of customers. Also, the right time marketing can optimize revenues in the seasons.

7.4 GEOGRAPHIC STRATEGY

The analysis by country showed that the **United Kingdom generates most of the total revenue**, which means that it performs well in one location but also, there is a high degree of geographic concentration.

Recommended Actions:

- Enter into **new foreign markets** to diversify the revenue lines.
- Enhance performance in **low performing areas** by launching specific campaigns.

- Regionalize products and marketing plans according to **regional tastes and behaviour**.
- Consider joint ventures or local sales in the new markets.

Foreseen Effect: Geographic diversification will decrease the reliance on one market and provide growth prospects. It also aids in the creation of a more balanced and sustainable revenue structure.

7.5 INVENTORY AND DEMAND PLANNING

The monthly revenue trend analysis showed that there were definite **seasonal demand trends** whereby revenue was high in some months. This has direct consequences on the inventory and supply chain management.

Recommended Actions:

- Raise inventory levels when there is **high demand** so as to satisfy the customer demand.
- Efficient **logistics and supply chain management** to deal with peak volumes. Reducing excess inventory when they are **not in high demand** helps to minimize holding costs.
- Take advantage of past data to enhance **demand forecasting and planning**.

Projected Effect: The improved inventory control will result in a higher level of customer satisfaction, lower costs of operation, and more efficiency in the supply chain operations.

7.6 RISK AND BUSINESS SUSTAINABILITY

A number of possible risks of influencing long-term business performance were also pointed out during the analysis. The main causes of these risks are the concentration in the products, customers and geographic markets.

Key Risks Identified:

- **Risk of product concentration** where the revenue is concentrated in a small number of products.
- **Risk of customer concentration**, in which few customers contribute to high proportion of the revenue.
- **Geographic concentration risk:** This is a situation in which a single market controls the overall sales.

Recommended Actions:

- Differentiate the product line with new or better products.
- Grow and attract a wider range of customers.
- Expand to new geographic markets to lessen reliance on one region.

Projected Impact: By mitigating these risks, it will contribute to a more resilient and sustainable business model, being less susceptible to market fluctuations and being able to maintain itself over time.

7.7 OVERALL STRATEGIC IMPLICATION

The general analysis reveals that the factors affecting the revenue generation of an e-commerce business are more than just a few factors which are linked to each other such as the product performance, customer behaviour, geographic distribution, and seasonal trends.

Using these insights, companies are able to transform reactive decision-making towards a more **proactive and data-driven approach** that allows companies to better allocate resources, enhance customer interactions, and increase revenue over the long term.

8. CONCLUSION

The aim of this project was to analyse a massive e-commerce transactional dataset in order to determine the most important drivers of revenue, and to learn the business performance trends. Through techniques of data analytics, the study analyzed various aspects of the data, such as revenue distribution, product performance, customer contribution, geographic sales trends, and seasonal trends.

It was found that the **distribution of revenue is not equal**, but it is concentrated in a few products and customers. There is a small number of high-performing products which bring a large share of the total revenue and a small number of high-value customers who is very important in making an overall sale. Also, the analysis by country showed that there was a high reliance on the **United Kingdom**, which implies geographic concentration.

The research also found that there were evident **seasonal trends** in revenue as sales rose during certain seasons especially towards the end of the year. This underlines the need of time based planning in other areas like inventory management and marketing strategy. More so, the Average Order Value (AOV ≈ 480.76) shows that there are medium spending customers per order, which means that by implementing the bundling and cross-selling strategies, the business can raise its revenues.

Among the most important lessons of this project is the need to employ data analytics to guide business decision-making. Businesses no longer have to make assumptions; data can be used to establish trends, measure performance, and create specific strategies. The lessons learned through this analysis offer valuable lessons on how to enhance product strategy, customer interaction, marketing performance, and geographic growth.

Comprehensively, the project will illustrate how **data analytics can convert raw transactional data into business insights that can be acted upon**. Through a systematic method of analysis, the research reveals the importance of using data in making decisions within the e-commerce industry. These could guide companies to maximize their revenues, enhance efficiency of their operations, and generate sustainable growth in a competitive world.